

VILLYAN SUTANTO

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SUMMARY

Computer Science and Mathematics undergraduate at BINUS University who builds end-to-end systems across machine learning, spatial analysis, optimization, and IoT. Experienced in developing YOLOv8-based traffic optimization systems and PostGIS-backed WebGIS platforms. Seeking internships in AI Engineering, Data Analysis, or GIS Engineering.

EDUCATION

Bina Nusantara (BINUS) University - School of Computer Science 2023 - 2027
Bachelor's degree in Computer Science and Mathematics • **GPA 3.85/4.00** Jakarta, Indonesia
Relevant coursework: Computational Geometry, Abstract Algebra, Real Analysis, Mathematical Modelling, Cloud Architecture (AWS), Data Structures, Algorithm Design, Software Engineering.

EXPERIENCE

Regional Secretary - Core Governing Board Feb 2025 - Feb 2026
Keluarga Mahasiswa Katolik (KMK) BINUS University @Kemanggisian Jakarta, Indonesia

- Managed incoming and outgoing organizational documents such as proposals, accountability reports, MoA, MoU, Terms of Reference, and Minutes of Meeting.
- Monitored and evaluated the performance of division heads and their teams in executing departmental responsibilities and organizational events.

Freshmen Mentor — Freshmen Leader and Freshmen Partner Aug 2024 - Feb 2026
First Year Program B28, BINUS University @Kemanggisian Jakarta, Indonesia

- Guided new students through the transition to campus life, introducing the university environment, communities, activities, academic programs, and available resources.
- Mentored freshmen on BINUSian values and the Sustainable Development Goals, and supervised their community service activity on sustainable energy awareness in schools.

PROJECTS

StaSIUN - Station Spatial Intelligence for Urban Network 2026 - Present
Most Favorite WebGIS Winner MAPID WebGIS Competition 2026 · github.com/Vleeflu/StaSIUN · Next.js · FastAPI · PostGIS · QGIS · OCR · NLP

- Built a geospatial economic intelligence platform that scores 79 Jakarta KRL commuter-rail stations on a five-variable Spatial Economic Potential Index (SEPI), using walking isochrones and ~32k POIs. Set up a FastAPI + Next.js monorepo on a Dockerised PostGIS database with Alembic migrations.
- Weighted SEPI with a hybrid Entropy + AHP scheme ($CR < 0.10$), quantified uncertainty with BCa bootstrap intervals and shrinkage, and ran a sensitivity analysis on the rankings. Designed a naming-rights valuation using TOPSIS ranking, benchmarked against MRT/LRT stations whose naming rights have already been sold, plus an LLM assistant with tool calling over the scoring API.

Traffic Signal Optimization with Computer Vision and Metaheuristics May - June 2026
Applied Mathematics Modelling and Undergraduate Thesis Research · github.com/Vleeflu/TrafficSignalOptimizationHHO · Python · YOLOv8 · OpenCV · Harris Hawks Optimization

- Integrated YOLOv8 vehicle detection with Markov queueing simulation and empirical arrival distributions to optimize signal phase timings, replacing the Genetic Algorithm baseline with Harris Hawks Optimization to accelerate convergence during phase-timing search.
- Processed ~2.9M detection records, converted vehicle counts to PCU per PKJI 2023, and dynamically calibrated saturation flow against a $\rho = 0.80$ target, achieving a 49.9% reduction in traffic delay.

Gojek App Review Sentiment Analysis May - June 2026
Text Mining and NLP Course Project · github.com/Vleeflu/GojekSentimentReview · Python · IndoBERT · Topic Modeling

- Classified sentiment in Indonesian-language Gojek reviews by application version with IndoBERT and surfaced complaint themes with LDA; co-authored an IEEE-format report.

Integrated IoT Door Lock with RFID, Ultrasonic Sensor, and Camera May - June 2025
Embedded Systems and IoT Course Project · github.com/Vleeflu/IoTDoor_RFID_Ultrasonic · ESP32 · Arduino IDE · C++

- Developed a smart door-lock prototype on an ESP32 combining RFID authentication, ultrasonic proximity detection, and camera-based monitoring, with control logic in C++.

SKILLS AND OTHER ACHIEVEMENTS

- Programming:** Python, C++, SQL, PHP, JavaScript, TypeScript, HTML/CSS
- ML, NLP, Computer Vision:** TensorFlow, Keras, PyTorch, YOLOv8, BERT, Word2Vec
- Geospatial:** PostGIS, QGIS, Leaflet.js, spatial clustering, Voronoi / Delaunay methods
- Optimization & Decision:** AHP, SAW, TOPSIS, ILP, dynamic programming, Markov chains, metaheuristics (Harris Hawks Optimization, GA)
- Engineering:** FastAPI, Next.js, Docker, Git / GitHub, pytest (TDD), Vercel, AWS (EC2, S3, Lambda)
- Certificate & award:** Fundamentals of Deep Learning (NVIDIA) · Top 50, MAPID WebGIS Competition 2026
- Spoken languages:** Indonesian (native) · English (professional working)